Systems of linear equations

1

a 1

b 0

c 1

d Infinite

e 0

2

a

i One solution

ii $(5, -1)$

b

i One solution

ii $(-6, 4)$

c

i One solution

ii $(2, -4)$

d

i One solution

ii $(3, -6)$

e

i One solution

ii $(6, -3)$

f

i One solution

ii $(-4, -2)$

3 No because the gradient can be anything.

4

a To intersect at $(0, 3)$ the line must have a y-intercept of 3, therefore $b = 3$.

b The value of $m \neq 5$. If $m = 5$ and $b = 3$ then the second line is the same as the first and there is an unlimited number of solutions.

5

a Yes

b No

c No

d No

Solve systems graphically

6

a $x = 6, y = 4$

b $x = 0, y = -2$

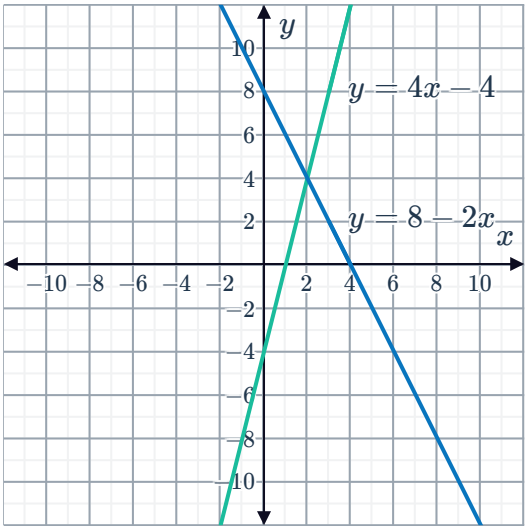
c $x = 4, y = 0$

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7

a

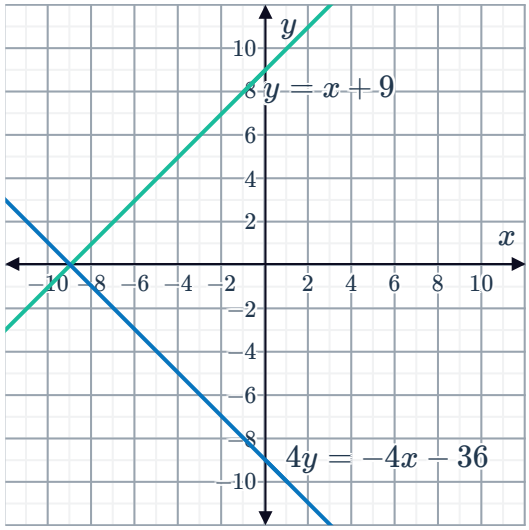
i



ii $(2, 4)$

b

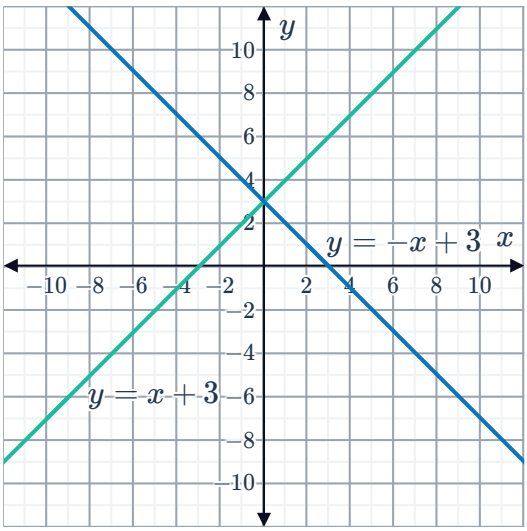
i



ii $(-9, 0)$

c

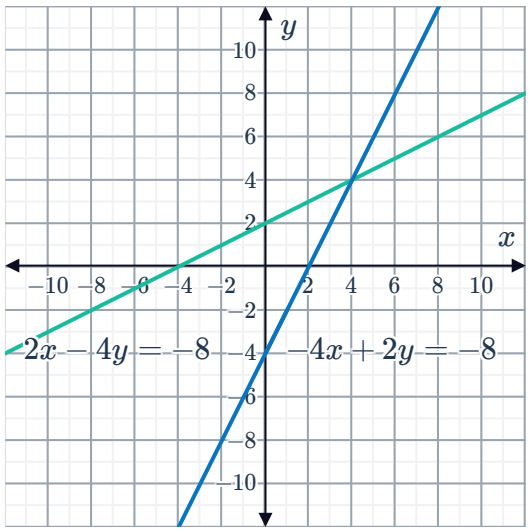
i



ii $(0, 3)$

d

i

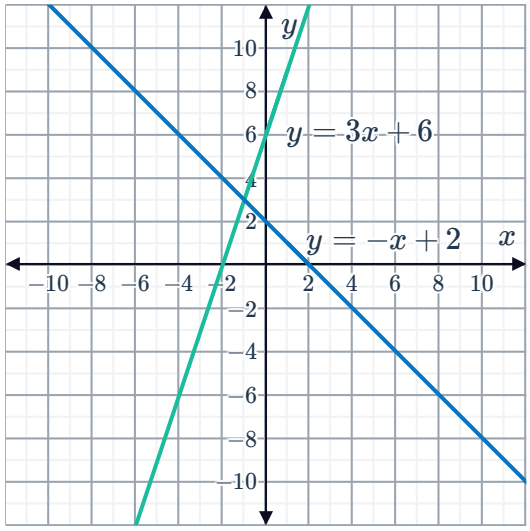


ii $(4, 4)$

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e

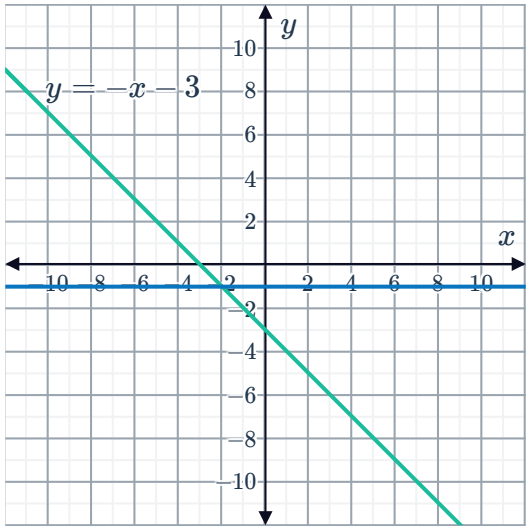
i



ii $(-1, 3)$

f

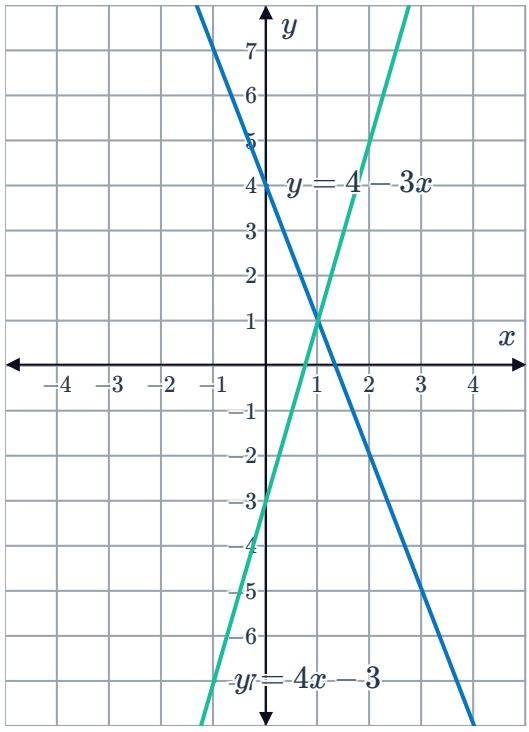
i



ii $(-2, -1)$

g

i

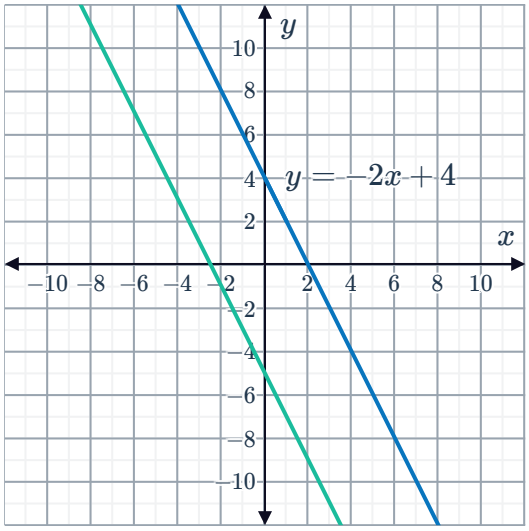


ii $(1, 1)$

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8

a



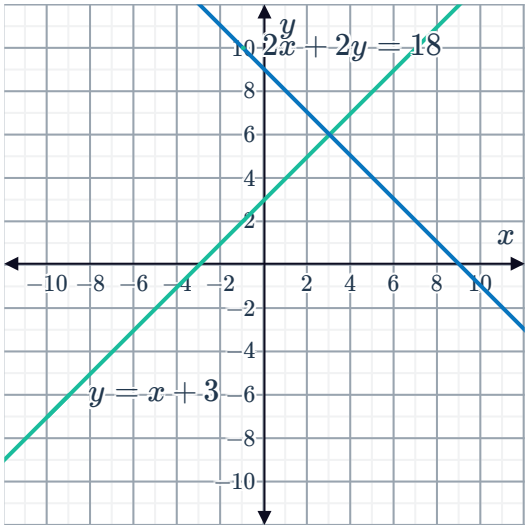
b No, because the lines are parallel.

Applications

9

a $y = x + 3, 2x + 2y = 18$

b

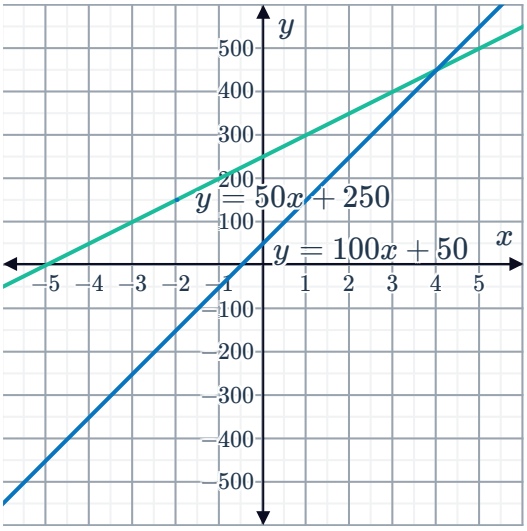


c 6 m and 3 m

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10

a

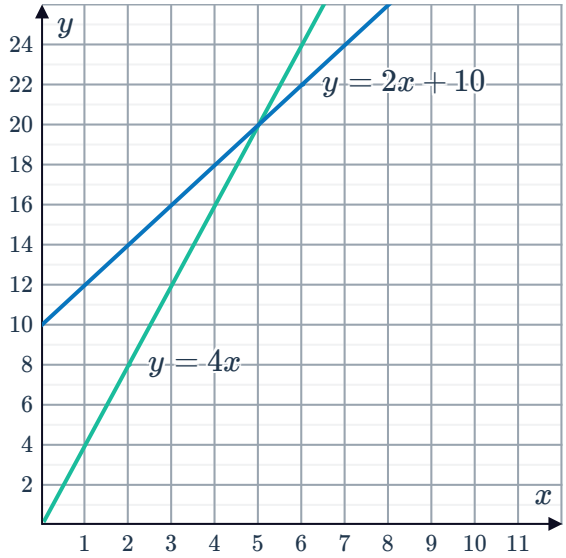


b $(4, 450)$

c Renting either studio A or B for 4 hours costs the same amount, $\$450$.

11

a



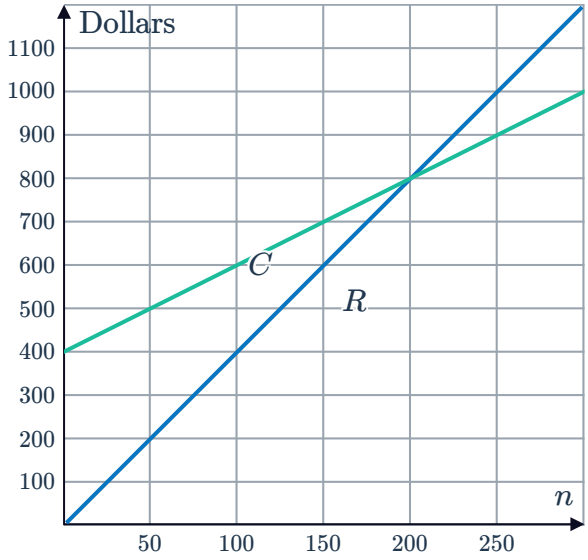
b $(5, 20)$

c Nonmembers and members pay the same amount per month for taking 5 classes, namely $\$20$.

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12

a



b 200

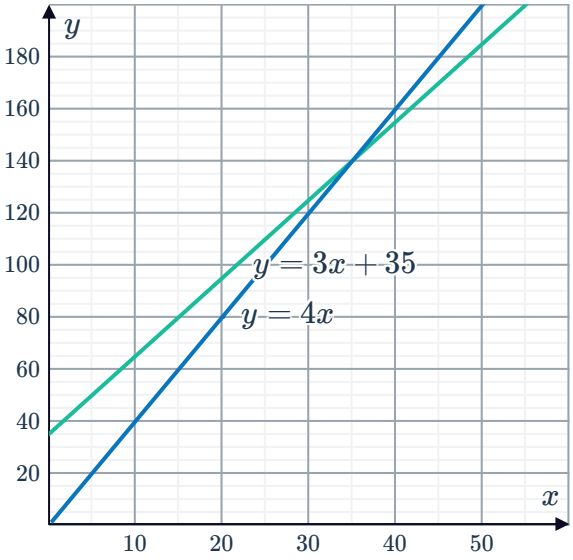
c The amount of revenue generated at the break-even point.

13 $(775, 2325)$

14

a $(35, 140)$

b



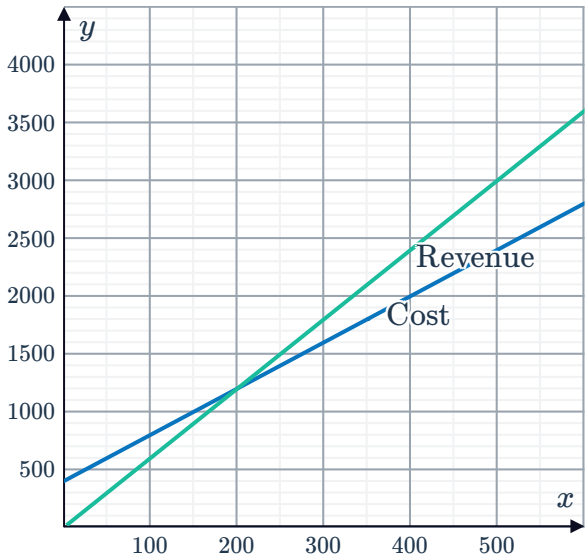
c The point at which Laura is able to just cover her expenses.

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15

a $(200, 1200)$

b



c The company's break even point, at which its revenue is equal to its costs.

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